

Public Economics (ECON 131)

Section #8: Externalities (EXTRA)

GSI: Steve Ramos

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1.1 When Marginal Damage is Not Constant

A coal-fired power plant releases air pollution into the atmosphere for every unit of electricity produced. The inverse demand function for coal-fired electricity is $P_d = 20 - \frac{1}{2}Q$, which represents the marginal benefit curve where Q is the quantity consumed when consumers pay price P_d . The inverse supply curve for coal-fired electricity is $P_s = 5 + Q$, which represents the marginal private cost curve when the power plant produces Q units. The marginal damage from emissions is given by $MD = 3.5Q$, which describes the cost of greenhouse gas emissions and local air pollution when the industry generates Q units of coal-fired electricity.

- (a) Illustrate the market for the coal fired electricity with a supply/demand graph. Be sure to draw the curves for demand, supply, marginal damage, and social marginal cost.

(b) What are the equilibrium price and quantity for coal fired electricity when there is no correction for the externality?

(c) How much coal fired electricity should the market supply at the social optimum?

(d) How large is the deadweight loss from the externality?

- (e) Is it possible for the government to achieve the social optimum by imposing a per-unit fee on emissions? If not, explain why it is not possible. If so, how large must the emission fee be if the market is to produce the socially efficient amount of coal fired electricity? Also, draw the firm's supply curve with the new emission fee on your graph.